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(54) MIXED PRECISION CAPABLE HARDWARE FOR TUNING A MACHINE LEARNING MODEL

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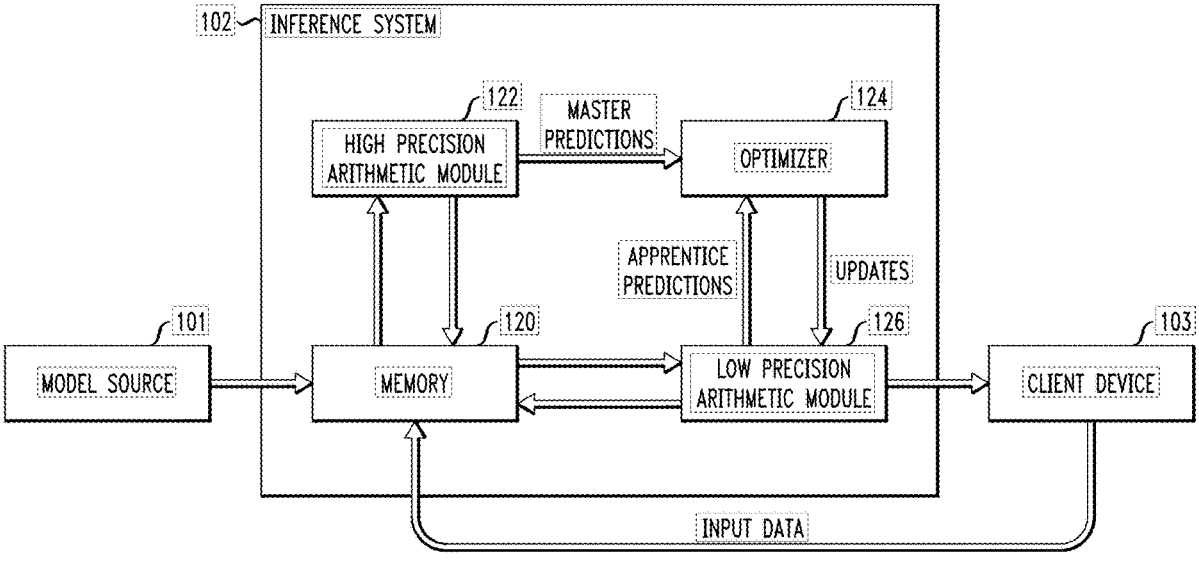
(57) **ABSTRACT**

An apparatus includes a memory and a processor coupled to the memory. The processor includes first and second sets of arithmetic units having first and second precision for floating-point computations, the second precision being lower than the first precision. The processor is configured to obtain a machine learning model trained in the first precision, to utilize the second set of arithmetic units to perform inference on input data, to utilize the first set of arithmetic units to generate feedback for updating parameters of the second set of arithmetic units based on the inference performed on the input data by the second set of arithmetic units, to tune parameters of the second set of arithmetic units based at least in part on the feedback generated by the first set of arithmetic units, and to utilize the second set of arithmetic units with the tuned parameters to generate inference results.